



Ilian SEBTI

EMBEDDED SYSTEMS ENGINEER

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PROFESSIONAL SUMMARY

Embedded systems engineer (ESILV, class of 2026) with hands-on experience across hardware and software: FPGA development in Verilog, embedded firmware on ESP32/Arduino, RF (licensed amateur radio operator, ANFR 2025), and full-stack tooling in Python and C++/C#. Built a real-time video synthesizer on an iCE40 FPGA and an EEG-controlled robot combining embedded firmware with real-time signal classification. Took embedded and data tools from prototype to field deployment during two internships at Vallourec. Electrical engineering exchange semester at Hanyang University (Seoul).

TECHNICAL SKILLS

Languages: Verilog (FPGA), C++, C#, Rust, Python, SQL, HTML/CSS, JavaScript

Embedded & Hardware: ESP32 / Arduino, KiCad, circuit design & soldering, RF (ANFR-licensed)

Software & Tools: Qt (PyQt5/PySide6), Git, REST API, Databricks / ETL pipelines, SCRUM

CAD: SolidWorks, Fusion 360

Languages spoken

French: Native

English: C1

Spanish: B1

PROJECTS

FPGA Video Synthesizer | 2025

Real-time video synthesizer built on an iCE40 FPGA in Verilog, generating and manipulating video signals at the hardware level.

Mind Controlled Robot | 2024

EEG-based brain-computer interface: Muse 2 headband to an ESP32 RC car via real-time EEGNet classification, WebSocket command layer, live ESP32-CAM video stream with VR viewing.

Bioprinter Custom Extrusion Head | 2025

Syringe-based bioprinting head for the Cosmyx NOVA 3D printer, designed and modeled in Fusion 360.

EXPERIENCE

Vallourec | Application Development & Data Science Intern

May – Sept 2025 & Feb – Aug 2026 | Meudon, France

- Designed and built multiple GUI application from scratch for PC and mobile for verifying pipe connector installations on oil rigs; developed and distributed 20+ versions to field technicians.
- Built ETL pipelines on Databricks feeding an internal data visualization tool; automated data cleaning and preprocessing workflows.
- Collaborated cross-functionally on app design, security, and deployment; incorporated field-technician feedback into the release cycle.
- Stack: Python, Qt, Databricks, Streamlit, Pandas/NumPy, OpenCV, Rust, Javascript/Typescript

Canal+ International | Web Development Intern (Backend)

June – Aug 2023 | Paris, France

- Developed and tested REST API endpoints using Django REST Framework and Swagger.
- Worked in an agile SCRUM team with Git for version control, participating in daily stand-ups.

EDUCATION

ESILV | Léonard de Vinci Engineering School — 2021 – 2026, Paris La Défense, France

General Engineering Degree

Hanyang University – Seoul, South Korea — Aug – Dec 2023

Exchange semester in Electrical Engineering

Lycée Alexandre Dumas – Saint-Cloud, France — June 2020

High School Diploma, honors

CERTIFICATIONS & INTERESTS

Certifications: Amateur Radio (ANFR, 2025), Design Thinker (Dthinking Academy, 2022) | **Interests:** Karate (black belt), Public Speaking Club (ESILV)

Soft Skills: Organization, critical thinking, teamwork, adaptability, curiosity, creativity